

Walmart: Big Data analytics at the world's biggest retailer

With over 20,000 stores in 28 countries, Walmart is the largest retailer in the world. So it's fitting then that the company is in the process of building the world's largest private cloud, big enough to cope with 2.5 petabytes of data every hour. To make sense of all this information, Walmart has created what it calls its Data Café – a state-of-the-art analytics hub located within its Bentonville, Arkansas headquarters.

How Walmart uses Big Data in practice

The Data Café allows huge volumes of internal and external data, including 40 petabytes of recent transactional data, to be rapidly modelled, manipulated and visualised. Speaking to me about the project, senior statistical analyst Naveen Peddamail said, "If you can't get insights until you've analysed your sales for a week or a month, then you've lost sales within that time."

Quick access to insights is therefore vital. For example, Peddamail told me about a grocery team who could not understand why sales had suddenly declined in a particular product category. By drilling into the data, they were quickly able to see that pricing miscalculations had led to the products being listed at a higher price than they should have been.

The system also provides automated alerts, so, when particular metrics fall below a set threshold in any department, the relevant team is alerted so that they can find a fast solution. In one example of this, during Halloween, sales analysts were able to see in real time that, although a particular novelty cookie was very popular in most stores, it wasn't selling at all in two stores. The alert prompted a quick investigation, which showed that, due to a simple stocking oversight, the cookies hadn't been put on the shelves. The store was then able to rectify the situation immediately.

The technical details

As well as 200 billion rows of transactional data (representing only the past few weeks!), the Café pulls in information from 200 sources, including meteorological data, economic data, Nielsen data, telecom data, social media data, gas prices, and local events databases. Anything within these vast and varied datasets could hold the key to the solution to a particular problem, and Walmart's algorithms are designed to blaze through them in microseconds to come up with real-time solutions.

Ideas and insights you can steal

Clearly, Walmart has huge amounts of data at its fingertips – and the resources to tackle all that data. But what any company can borrow from Walmart's example is their ability to react to data quickly. After all, there's little point investing in data capabilities if your internal setup doesn't allow you to quickly make decisions and

changes based on what the data is telling you.

You can read more about how Walmart is using Big Data to drive success in [Big Data in Practice: How 45 Successful Companies Used Big Data Analytics to Deliver Extraordinary Results](#).



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